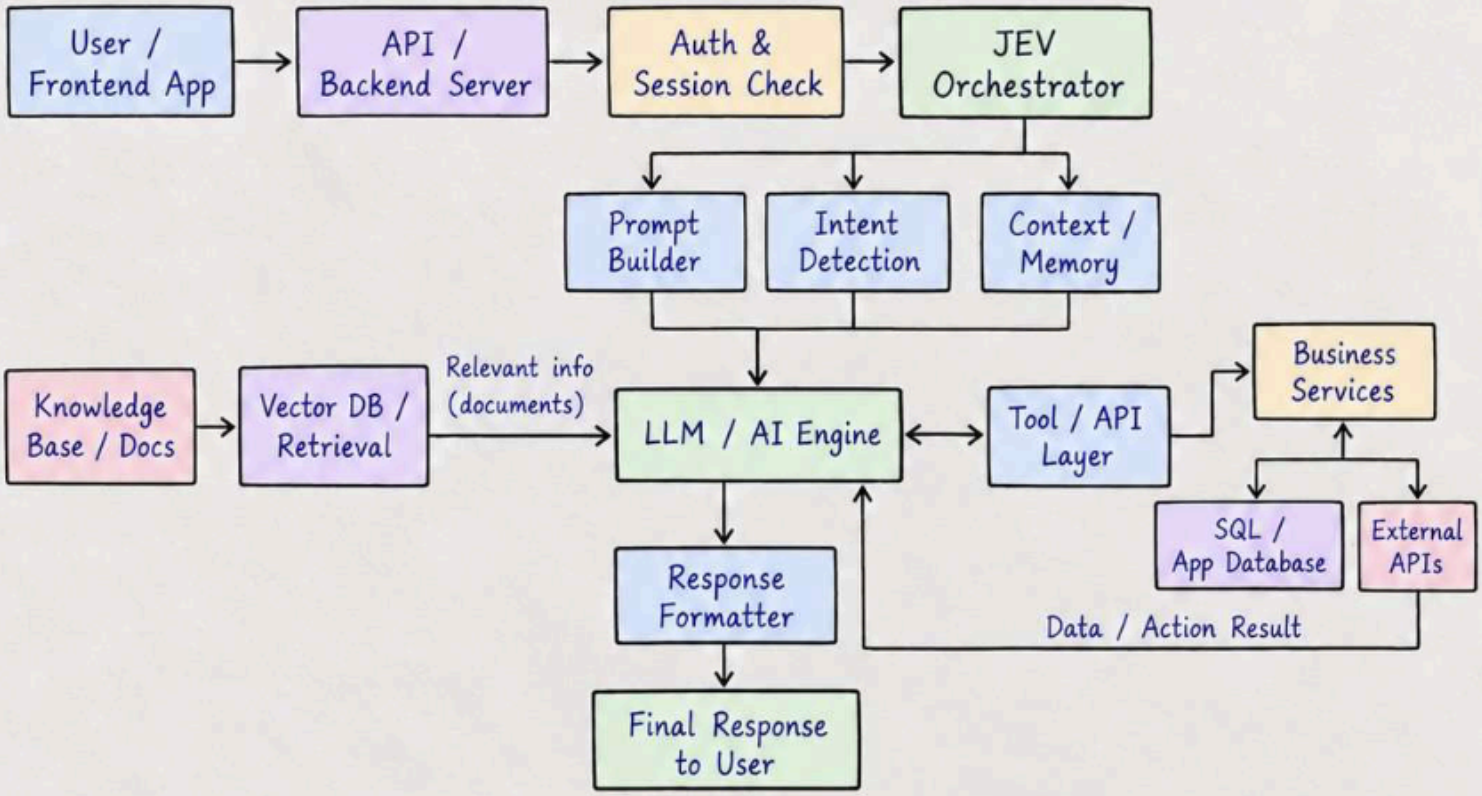
JEV Model

1. Complete Backend Architecture



2. Main Flow

1. User sends a natural language request.
2. Backend authenticates and forwards the request.
3. JEV understands the intent.
4. Relevant context and documents are retrieved.
5. LLM thinks and may call tools if needed.
6. Tools fetch data from databases or external APIs.
7. Response is formatted into a useful answer.
8. Final response is returned to the user.

3. Why this architecture works

- Handles natural language requests.
- Uses memory and retrieval for better accuracy.
- Can perform real backend actions through tools.
- Connects AI with databases and services.
- Returns structured, useful responses.

4. Mini Example

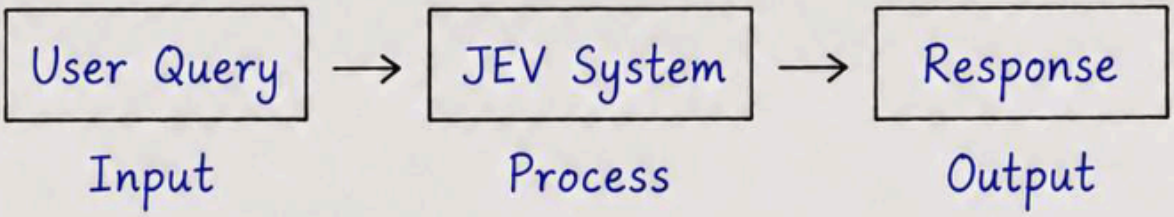


JEV Model

1. What is JEV Model?

- JEV Model is a simple AI workflow model.
- It helps a system understand a user request and generate a useful response.
- It combines input, intelligence and action.

2. Basic Flow



3. Why it is used

- Understands natural language.
- Gives smarter responses than fixed-rule systems.
- Can connect with tools and data.

4. Simple Example

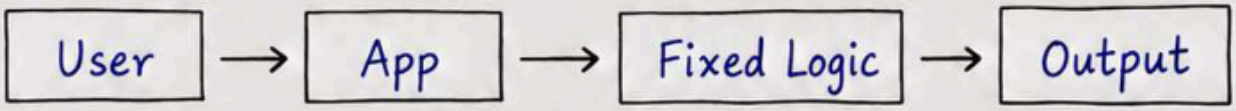
User: "What is weather in Delhi?"

→ JEV reads request → returns answer.

JEV Model

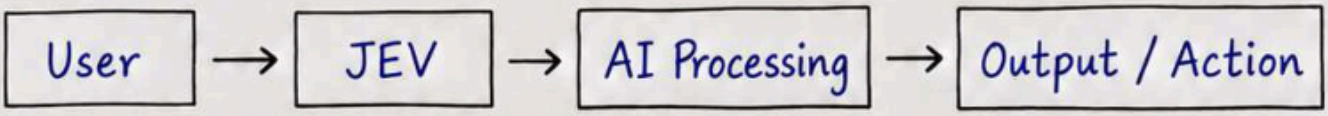
1. Traditional System

- Works on fixed rules.
- Good for predefined flows.
- Struggles with natural language.



2. JEV-Based System

- Accepts natural language.
- Understands user intent.
- Can use context and tools.



3. Quick Comparison

Traditional System	JEV-Based System
Works on fixed rules.	Accepts natural language.
Limited flexibility.	Flexible.
Less context.	Context-aware.

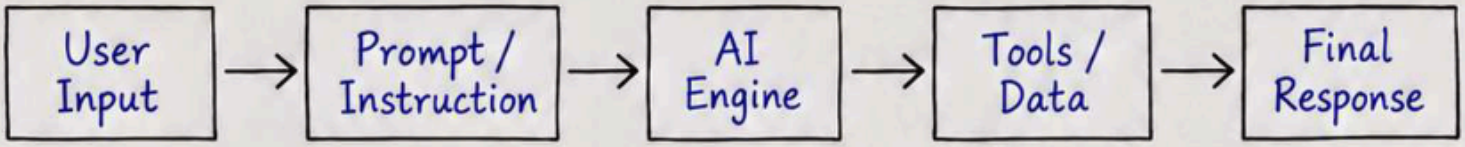
JEV is useful when user requests are dynamic.

JEV Model

1. Core Components

- **User Input** : The question, request or information provided by the user.
- **Prompt / Instruction** : The input formatted with clear instructions, context or rules.
- **AI Engine** : The core model that processes the input, reasons and decides the next steps.
- **Tool / API Layer** : External tools or APIs used to fetch real-time information, perform actions or run code.
- **Knowledge / Data** : Internal knowledge, training data or external data sources used for understanding and answering.
- **Final Response** : The complete and useful answer returned to the user.

2. Main Structure



3. Remember

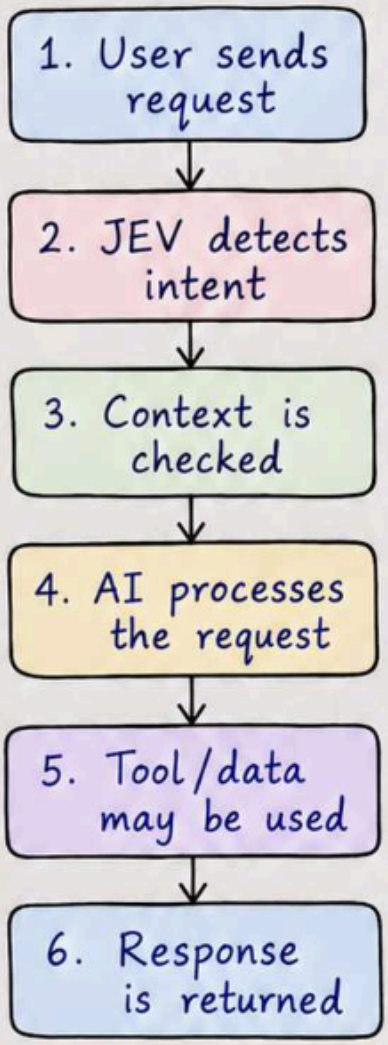
- JEV is not just a model response.
- It can read, think, fetch and answer.

Each component improves the final answer quality.

JEV Model

1. How JEV Works

1. User sends request.
2. JEV detects intent.
3. Context is checked.
4. AI processes the request.
5. Tool/data may be used.
6. Response is returned.



2. Mini Example

User asks: "Summarize this PDF"

Input → "Summarize this PDF"

Understand → JEV detects the intent (summarization).

Read context → JEV checks the PDF content and relevant context.

Generate answer → JEV creates a clear and concise summary.

JEV works step-by-step, not by random guessing.

JEV Model

1. Prompt

A prompt is the input instruction given to the system.
Better prompt = better output.

2. Intent

Intent means what the user actually wants.
Example: asking, searching, summarizing, booking.

3. Context

Context gives supporting information.
It helps the answer become more accurate.

4. Example

Basic: "Explain Java"
Better with context: "Explain Java for interview preparation in simple language"

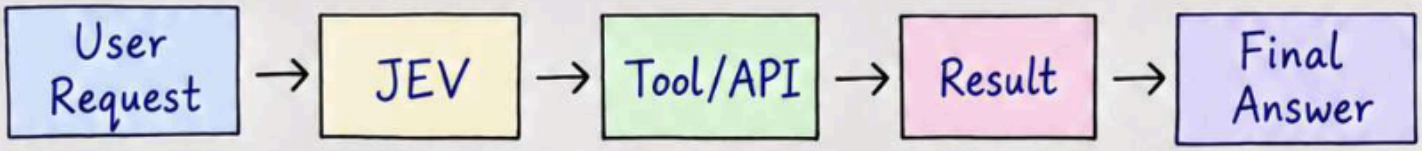
→ Prompt + intent + context = smarter responses.

JEV Model

1. Tool Calling

- JEV can use external tools when needed.
- This helps it do real tasks, not only text generation.

2. How it works



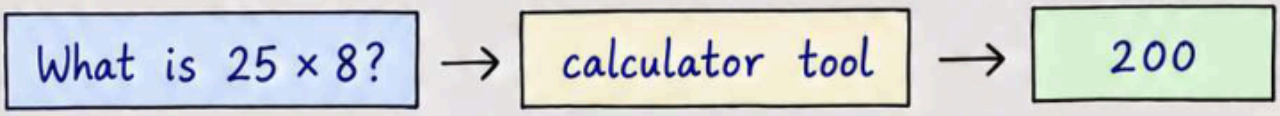
3. Examples

- Weather API
- Calculator
- Search tool
- Database query

4. Why useful

- Gives real-time or exact results.
- Expands system capability.

5. Example



JEV Model

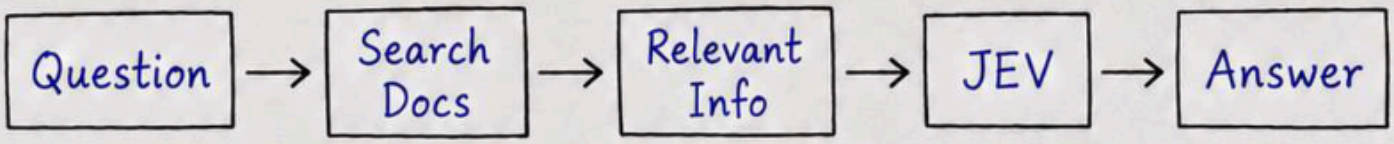
1. Knowledge vs Retrieval

- Model knowledge = what the system already knows.
- Retrieval = fetching external information when needed.

2. Memory

- Memory helps keep conversation continuity.
- It remembers earlier user context.

3. Retrieval Flow



4. Why important

- Improves accuracy.
- Helps answer domain-specific questions.
- Useful in document Q&A systems.

This is the base idea behind knowledge assistants and RAG-like systems.

JEV Model

1. Output Types

- Plain text output
- Structured output

2. Why structured output?

- Easy to parse
- Useful in apps and automation
- More predictable response format

3. JSON Example

```
{
    "task": "book_flight",
    "city": "Delhi",
    "date": "Tomorrow"
}
```

4. Use Cases

- form filling
- APIs
- agents
- workflow automation

Structured outputs help software systems consume AI responses more reliably.

JEV Model

1. Common Limitations

- Hallucination
- Missing context
- Wrong tool usage
- Incomplete or vague prompt

2. Guardrails

- Validate output
- Restrict sensitive actions
- Check tool responses
- Use human review when needed

3. Problem → Fix

Problem	Fix
Hallucination	verify facts
Vague prompt	add details
Tool error	retry / check result

Smart systems need checks, not just intelligence.

JEV Model

1. Real Use Cases

- Customer support bot
- Coding assistant
- AI search assistant
- Knowledge assistant
- Workflow automation

2. Quick Revision

- takes input
- understands intent
- uses context
- may call tools
- returns response

3. Simple Full Flow



JEV Model connects understanding, reasoning and action.